

Robb Calder

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Education

PhD in Astrophysics

Institute of Astronomy, University of Cambridge

Cambridge, UK

Sep 2023 – Present

- Funded studentship from the Science and Technology Facilities Council, UKRI.
- Project title: ‘Atmospheric Processes on Hot Rocky Planets’.
- Supervisors: Prof. Oliver Shorttle and Dr. Paul Rimmer.

Integrated MPhys in Astrophysics

School of Physics and Astronomy, University of St Andrews

St Andrews, UK

Sep 2018 – Jun 2023

- Graduated First Class Honors with Distinction; 83/100.
- Carried out a research project under the supervision of Professor Moira Jardine titled *Smothering Exoplanet Winds*.
- Studied a wide range of courses including *Advanced Data Analysis*, *Magnetofluid Dynamics*, *Observational Astrophysics*, *Monte-Carlo Radiative Transfer Techniques* and *Observational Astrophysics*.
- Awarded class medal in final year for being among the highest ranking astrophysics students.
- Deans Scholar for each consecutive year of undergraduate degree.

Research Experience

PhD - Atmospheric Processes on Hot Rocky Planets

Institute of Astronomy, University of Cambridge

Cambridge, UK

Sep 2023 – Present

- Performed retrievals on transmission spectra of K2-18b and WASP-17b using the petitRADTRANS and PLATON retrieval packages.
- Modeled the atmospheres of Venus and Venus-like exoplanets with a 1D photochemical code (ARGO) to discover possible abiotic ozone production pathways.
- Modified ARGO to include pseudo-chemical reactions that produce arbitrary amounts of specific gaseous species to simulate unknown chemical pathways.
- Simulated the thermal evolution of planets in the sub-Neptune regime across a broad parameter space using a coupled atmosphere-interior evolution model (PROTEUS).
- Contributed to the collaboration responsible for developing PROTEUS by attending regular meetings and sharing changes made to the model to the collaboration’s github repository.
- Tested the magma ocean hypothesis for K2-18b by comparing PROTEUS simulations of this planet to observed transmission spectra.
- Added a 0D mantle evolution sub-module within PROTEUS with parameterised convection, to be used in planet mass regimes unsuitable for the more sophisticated 1D mantle evolution sub-module.
- Incorporated petitRADTRANS into the PROTEUS framework, allowing the code to produce synthetic observations based on model results.

Masters’ Project - Smothering Exoplanet Winds

School of Physics and Astronomy, University of St Andrews

St Andrews, UK

Jan 2023 – May 2023

- Used a 1D Parker Wind model to determine if stellar winds act to reduce the photo-evaporation rate of a planetary atmosphere.
- Determined the region of the parameter space; defined by stellar mass, planet mass and orbital separation; in which the stellar wind reduces the atmospheric evaporation rate.
- Explored the effect of stellar flares and stellar evolution on the ability of stellar winds to reduce atmospheric

evaporation rates.

- Identified that the planet HD 97658b is expected to have a reduced atmospheric evaporation rate owing to the stellar wind, which may explain the lack of an observed evaporating atmosphere.

Summer Research Project - Cloud Formation on Hot Jupiter Exoplanets

St Andrews, UK

School of Physics and Astronomy, University of St Andrews

Jun 2021 – Oct 2021

- Received funding from the Laidlaw Leadership and Research Scholarship to undertake a research project under the supervision of Prof. Christiane Helling.
- Studied cloud formation on Hot-Jupiter exoplanet atmospheres using a grid of Global Circulation Models as input for a 1D kinetic cloud formation code.
- Visualised the GCM data using Python to inform the cloud formation simulations and identify pressure-temperature regimes that may prove unsuitable for the model.
- Created a bash script to automate the distribution of cloud formation simulations across a high performance computing cluster.
- Designed a strategy for visualising the vast quantity of high-dimensional output data (60GB) in a concise and insightful format.

Selected Publications

- Calder, R., Shorttle, O., Nicholls, H., Lichtenberg, T., & Guimond, C. M. (2026). Most rocky sub-Neptunes are molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets. *Monthly Notices of the Royal Astronomical Society*, 549(3), stag1007.
- Calder, R., Shorttle, O., Jordan, S., Rimmer, P., & Constantinou, T. (2025). Abiotic ozone in the observable atmospheres of Venus and Venus-like exoplanets. *Monthly Notices of the Royal Astronomical Society*, 540(3), 2432-2450.
- Helling, C., Samra, D., Lewis, D., Calder, R., Hirst, G., Woitke, P., ... & Chubb, K. L. (2023). Exoplanet weather and climate regimes with clouds and thermal ionospheres-A model grid study in support of large-scale observational campaigns. *Astronomy & Astrophysics*, 671, A122.
- Lewis, D., Helling, C., Calder, R., Hirst, G., Samra, D., Woitke, P., ... & Carone, L. (2022). The supplementary catalogue: Exoplanet weather and climate regimes with clouds and thermal ionospheres.

Technical Skills

Programming

- Python (including numpy, matplotlib, astropy and scipy), Mathematica, Fortran, Bash.
- L^AT_EX, HTML, CSS.

Coding Packages

- Use and modification of 1D photochemical-kinetics code (ARGO) to simulated chemistry in the upper-atmosphere of Venus-like planets.
- Use of retrieval packages (petitRADTRANS and PLATON) for atmospheric retrievals and synthetic spectra generation.
- Use and modification of coupled atmosphere-interior evolution model PROTEUS to simulate the thermal evolution of planets in the sub-Neptune regime.

Conferences and Invited Presentations

IoA 50 - Frontiers Conference

Attended in Person

Cambridge, UK

July 2024

Origins Federation Conference

Attended in Person

Cambridge, UK

Sep 2024

Leverhume Centre for Life in the Universe Science Day, 2025

Attended in Person, presented a poster titled: 'Abiotic Ozone in the Observable

Cambridge, UK

February 2025

Atmospheres of Venus and Venus-like Exoplanets'

UK Exoplanet Meeting 2025

Attended in Person, presented a poster titled: 'Abiotic Ozone in the Observable Atmospheres of Venus and Venus-like Exoplanets'

Leeds, UK

March 2025

Biennial European Astrobiology Conference 2025

Attended in Person, presented a poster titled: 'Abiotic Ozone in the Observable Atmospheres of Venus and Venus-like Exoplanets'

Reykjavik, Iceland

June 2025

Exoplanets Seminar

Presented a talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Cambridge, UK

December 2025

Rocky Worlds Conference 2026

Attended in Person, presented a talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Groningen,

Netherlands

January 2026

Leverhume Centre for Life in the Universe Science Day, 2026

Attended in Person, presented a talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Cambridge, UK

February 2026

UK Exoplanet Meeting 2026

Attended in Person, presented a talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Bristol, UK

March 2026

Invited Talk at MIT

Presented invited talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Boston, USA

May 2026

Invited Talk at Harvard

Presented invited talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Boston, USA

May 2026

Invited Presentation at Astrobiology Science Conference 2026

Attended in Person, invited presentation titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Maddison, USA

May 2026

Exoplanets 6

Attended in Person, presented a poster titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Porto, Portugal

June 2026

Invited Talk at University of St Andrews

Presented invited talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

St Andrews, UK

July 2026

Invited Talk at University of Edinburgh

Presented invited talk titled: 'Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets'

Edinburgh, UK

July 2026

Teaching Experience

Undergraduate Supervisor

University of Cambridge

Cambridge, UK

Feb 2024 - May 2026

Directed sessions involving 2-3 students to help them consolidate material covered in lectures. Marked submitted worksheets and past paper questions.

- 1st Year: Topics in Astrophysics (3rd Year Undergraduate Course): 9 students, 12 sessions
- 2nd Year: Topics in Astrophysics (3rd Year Undergraduate Course): 4 students, 8 sessions

Outreach

Public Lectures and Stargazing

University of Cambridge

Cambridge, UK

Sep 2024 - May 2026

Delivered lectures to members of the public explaining current findings in the field of exoplanet science and led supervised star-gazing sessions.

- 21/10/2024: Barrington Scout Group
- 22/11/2024: 6th/17th Cambridge Scouts
- 17/02/2025: 50th Cambridge Cub Scouts
- 18/02/2025: Team Consulting

International Women's Day Committee

University of Cambridge

Cambridge, UK

Jan 2025 - Mar 2025

Participated in the International Women's Day committee. The committee organised various department activities with the aim of celebrating female astronomers and promoting gender diversity in the field of astronomy.

- Lead and organised a workshop for local secondary school children to teach them about exoplanet transits and famous female astrophysicists in the field of exoplanets.
- Helped run the International Women's Day stall at the Institute of Astronomy Open Day, which presented examples of female astronomers throughout history.
- Presented a lecture at the Institute of Astronomy Open Day about famous female astrophysicists in the field of exoplanet science.